PC-MTPNet With Prototype Constraints for Fine-Grained Rice Classification From Multi-Temporal Polarimetric SAR Images

Xianyu Guo, Junming Liang, Junjun Yin, *Senior Member, IEEE*, Kun Li, and Jian Yang, *Senior Member, IEEE*

***Abstract*—Multi-temporal polarimetric synthetic aperture radar (SAR) captures temporal dynamics and polarimetric scattering variations during the rice-growing season, providing critical information for crop growth monitoring and fine-grained rice classification. However, in complex SAR backgrounds, rice fields are often surrounded by water, shoals, and urban built-up areas, while speckle noise and mixed boundaries further hinder accurate fine-grained discrimination. Existing deep segmentation networks typically feed SAR features directly into the model, with limited explicit modeling of the statistical structure of multi-temporal polarimetric features and fine-grained rice categories. They also lack effective constraints on intraclass compactness and interclass separability, leading to substantial confusion between rice subcategories. To address these limitations, this study proposes PC-MTPNet for fine-grained rice classification using multi-temporal full- and compact-polarimetric SAR features. PC-MTPNet integrates the multi-temporal polarimetric statistical descriptor (MTP-Desc), the multi-temporal polarimetric prototype constraint (MTP-Polar), and a detail-refined U-Net++ backbone. A composite loss function is introduced to impose intraclass compactness and interclass separation, while an edge head, boundary loss, and Hann–Gaussian sliding-window fusion are employed to enhance boundary delineation and full-scene inference stability. Experimental results show that PC-MTPNet achieves an overall accuracy (OA), mean intersection over union (mIoU), macro-averaged *F*1 score (Macro-F1), and Kappa of 93.73%, 86.49%, 92.66%, and 92.11%, respectively. Compared with U-Net++, SegFormer-B1, CVNN-PolSAR, and other methods, PC-MTPNet improves these four metrics by at least 3.05, 5.11, 2.80, and 0.55 percentage points, with maximum gains of 11.80, 19.40, 12.83, and 14.72 percentage points, respectively. For direct-sown japonica rice paddy (D-J) and transplanted hybrid rice paddy (T-H), PC-MTPNet achieves classification accuracies of 93.50% and 92.49%, respectively, improving over the U-Net++ baseline by 17.57 and 16.50 percentage points, respectively. These results indicate that multi-temporal polarimetric statistical description and prototype constraints effectively reduce statistical confusion between rice subcategories and improve the robustness of fine-grained rice classification in complex polarimetric SAR scenes.**

***Index Terms*—Fine-grained rice classification, multi-temporal polarimetric statistical descriptor (MTP-Desc), multi-temporal polarimetric prototype constraint (MTP-Polar), multi-temporal polarimetric synthetic aperture radar (SAR), prototype constraint.**

I. Introduction

RICE is a major staple crop worldwide, and changes in its cultivated area, spatial distribution, and type composition are directly linked to food security, agricultural resource regulation, disaster monitoring, and crop yield assessment [1], [2]. For major rice-producing regions, timely and reliable rice planting information is essential for monitoring regional agricultural dynamics and provides remote sensing support for variety layout optimization, farmland management, and the promotion of high-yield cultivars. Optical remote sensing provides rich spectral and textural information and has been widely used for crop identification. However, during the rice-growing season in southern China and the East Asian monsoon region, persistent rainfall and cloud contamination often result in insufficient valid optical observations, limiting continuous monitoring of key rice phenological stages. Synthetic aperture radar (SAR), as an active microwave remote sensing technique, can acquire surface information under cloudy, rainy, and nighttime conditions [3], [4]. It is sensitive to surface structure, water status, and dielectric properties, making it particularly suitable for continuous observation in cloudy and rainy agricultural regions. Existing studies have extensively exploited the all-weather capability of SAR for rice mapping, crop type identification, and growth monitoring [5]. Nevertheless, most studies have focused on "rice–nonrice" discrimination or major land-cover classification, while fine-grained differences among rice planting types under complex SAR backgrounds remain insufficiently explored.

Compared with single-temporal SAR, multi-temporal SAR records backscattering variations across different growth stages, thereby reflecting phenological processes such as transplanting, regreening, jointing, heading, grain filling, and maturity [6]. Furthermore, polarimetric SAR, as well as compact-polarimetric SAR features simulated from fully polarimetric data, contains not only scattering intensity information but also differences in surface scattering, volume

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scattering, double-bounce scattering, channel-energy distribution, and polarimetric statistical structure [7], [8], [9], [10], [11]. Therefore, multi-temporal polarimetric SAR does not merely increase the number of input channels for rice identification; rather, it provides joint discriminative information from temporal evolution and scattering mechanisms. Previous studies have shown that multi-temporal fully polarimetric or compact-polarimetric SAR data can be used for rice growth monitoring and rice type classification [12], [13]. For example, compact-polarimetric SAR data have been used to analyze scattering intensity differences among rice types, indicating that polarimetric parameters at different phenological stages provide different discriminative capabilities for the two rice types. Other studies have constructed rice-height-sensitive indices and decision tree methods based on multi-temporal compact-polarimetric SAR, demonstrating that parameters such as surface scattering from m-χ decomposition and variations in Shannon entropy intensity contribute substantially to distinguishing different rice types [14]. These studies suggest that polarimetric parameters, phenological stages, and temporal scattering variations provide effective cues for fine-grained rice identification. However, most of them rely on manually designed indices, threshold selection, or decision tree rules, and have not transformed the multi-temporal polarimetric statistical structure into deep network constraints that can be optimized end-to-end.

Remote sensing-based rice identification has generally evolved from traditional machine learning and physical-feature modeling to deep-learning-based semantic segmentation. Traditional approaches typically use backscattering intensity, polarimetric decomposition parameters, vegetation indices, or handcrafted features, combined with classifiers such as RandomForest, support vector machine, or decision tree for land-cover recognition [15], [16], [17]. These methods offer a certain degree of interpretability and remain relatively robust when training samples are limited. However, their performance strongly depends on manual feature selection and threshold design, making it difficult to fully model spatial context in complex agricultural landscapes. Subsequently, fully convolutional networks, U-Net-like encoder–decoder architectures, and Transformer-based segmentation models such as SegFormer and SETR have become representative frameworks for semantic segmentation [18], [19], [20], [21], [22]. These deep models have also been widely adopted in remote sensing image interpretation and SAR-related agricultural applications [23], [24], [25]. Nevertheless, they are typically designed for general semantic segmentation tasks and do not explicitly model the phase–channel organization of multi-temporal polarimetric SAR. When six temporal phases and multiple polarimetric channels are directly stacked as ordinary multichannel inputs, the model may implicitly learn feature combinations, but the scattering evolution of rice across growth stages, polarimetric channel dispersion, and energy distribution differences are not explicitly represented. For fine-grained rice classification using multi-temporal polarimetric SAR, existing methods still face three main limitations. First, at the feature representation level, physical-feature-based methods can explain the discriminative effects of specific polarimetric parameters or temporal indicators, but they rely on manual selection and fixed rules. Existing deep segmentation networks can learn spatial context, but they usually treat SAR features as ordinary multichannel inputs, requiring the model to implicitly learn phenological changes and polarimetric statistical patterns from high-dimensional data, without an explicit multi-temporal polarimetric statistical representation. Second, at the class distribution level, existing methods mostly optimize pixelwise classification accuracy or regional overlap. Conventional softmax, cross-entropy, or region-based losses mainly constrain local predictions and lack explicit constraints on the intraclass compactness and interclass separability of D-J and T-H in the statistical space. Because D-J and T-H may exhibit similar backscattering intensities and field morphologies at certain phenological stages, relying only on pixelwise classification boundaries can easily lead to probabilistic confusion. Third, at the spatial mapping level, background classes such as water, shoal, and urban areas are often adjacent to rice fields in complex SAR scenes. Water edges, shoal transition zones, and urban built-up boundaries are vulnerable to speckle noise, mixed pixels, and sliding-window patch-edge effects [26], [27], resulting in fragmented field boundaries, increased local small patches, and discontinuous full-scene predictions.

To address these issues, this article proposes PC-MTPNet, a prototype-constrained framework for fine-grained rice classification from multi-temporal polarimetric SAR images. Unlike traditional machine learning methods that mainly rely on handcrafted polarimetric features and classifiers, PC-MTPNet preserves the original multi-temporal polarimetric SAR input and further constructs a physically interpretable multi-temporal polarimetric statistical description space. This description space explicitly characterizes temporal evolution, channel dispersion, energy distribution complexity, and adjacent-phase variations. In contrast to generic CNN- or Transformer-based segmentation networks that implicitly learn discriminative features from high-dimensional channels, multi-temporal polarimetric statistical descriptor (MTP-Desc) organizes the input of six temporal phases and eight polarimetric channels into an interpretable statistical basis for constraint learning.

Furthermore, to address the limitation of conventional pixelwise losses in constraining rice-subclass distributions, multi-temporal polarimetric prototype constraint (MTP-Polar) performs class prototype alignment in the MTP-Desc descriptor space and introduces margin separation between direct-sown japonica rice paddy (D-J) and transplanting hybrid rice paddy (T-H). This design enhances intraclass consistency and interclass separability between the two rice types. Finally, to alleviate fragmented field boundaries and discontinuous full-scene inference under complex SAR backgrounds, the detail-refined U-Net++ backbone is combined with an edge head, boundary loss, and Hann–Gaussian sliding-window fusion to improve boundary representation and the stability of full-scene probability maps.

The main contributions of this article are summarized as follows.

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1) PC-MTPNet is proposed for fine-grained rice classification from multi-temporal polarimetric SAR images under complex SAR backgrounds. Unlike general land-cover segmentation or rice/nonrice identification, the core task of this study is fine-grained discrimination between D-J and T-H rice paddies, while complex background classes are retained to evaluate model stability in real SAR scenes.
2) MTP-Desc and MTP-Polar are designed to improve the discriminative representation of rice subclasses. MTP-Desc constructs a physically interpretable multi-temporal polarimetric statistical description space from SAR inputs, enabling explicit representation of rice differences in temporal evolution, channel-energy distribution, channel dispersion, and polarimetric statistical structure. MTP-Polar performs class prototype alignment in this description space and introduces margin separation for rice subclasses, thereby enhancing intra-class compactness and interclass separation.
3) Boundary auxiliary supervision in the detail-refined U-Net++ backbone is integrated with Hann–Gaussian weighted sliding-window fusion to improve field boundary delineation and full-scene inference stability under complex backgrounds. This design enables the model to focus not only on pixelwise classification results but also on rice-subclass probability distributions, field boundary quality, and large-scale SAR mapping continuity. The effectiveness of PC-MTPNet is further validated through comparisons with multiple baseline methods.

II. Related Work

A. Theory of Polarimetric and Compact-Polarimetric SAR

The Sinclair polarimetric backscattering matrix, $S = [S_{HH} S_{HV}; S_{VH}\ S_{VV}]$, provides important characteristic parameters for describing the electromagnetic scattering properties of radar targets. In this study, three polarimetric features extracted from the Sinclair backscattering matrix, namely (S_{HH}, S_{HV}/S_{VH}, and S_{VV}), are used as part of the input feature set. In addition, compact-polarimetric SAR, as a dual-polarization SAR configuration, transmits microwave pulses in a single polarization mode and simultaneously receives (RVs) backscattered echoes in orthogonal polarization states. This design reduces system complexity and power consumption, increases imaging swath coverage, decreases sensor volume, and retains rich polarimetric information. Previous work has also shown that the combination of full-polarimetric (FP) and compact-polarimetric features outperforms the use of a single polarimetric mode for crop classification [28]. Therefore, in addition to FP parameters, compact-polarimetric parameters are also used as input features in this study.

For compact-polarimetric SAR, given an incident polarized wave, the four Stokes parameters can be derived from focused single-look complex data and expressed in a circular polarization basis as follows:

$$\begin{aligned} g_0 &= \left\langle |E_H|^2 + |E_V|^2 \right\rangle = \left\langle |E_L|^2 + |E_R|^2 \right\rangle \\ g_1 &= \left\langle |E_H|^2 - |E_V|^2 \right\rangle = 2\mathrm{Re}\left\langle E_L E_R^* \right\rangle \\ g_2 &= 2\mathrm{Re}\left\langle E_H E_V^* \right\rangle = 2\mathrm{Im}\left\langle E_L E_R^* \right\rangle \\ g_3 &= 2\mathrm{Im}\left\langle E_H E_V^* \right\rangle = \left\langle |E_L|^2 - |E_R|^2 \right\rangle \end{aligned} \quad (1)$$

where E_i denotes the unit Jones vector of the transmitted radar wave.

Taking right-circular transmission as an example, the Stokes vector g can be used to derive four compact-polarimetric backscattering coefficients under right-circular transmit and horizontal linear RV (RH), right-circular transmit and vertical linear RV, right-circular transmit and right-circular RV (RR), and right-circular transmit and left-circular RV (RL) configurations. The corresponding calculations are given in the following equations [29]:

$$\sigma^0_{RH} = [1, 1, 0, 0] \times g \quad (2)$$

$$\sigma^0_{RV} = [1, -1, 0, 0] \times g \quad (3)$$

$$\sigma^0_{RR} = [1, 0, 0, -1] \times g \quad (4)$$

$$\sigma^0_{RL} = [1, 0, 0, 1] \times g. \quad (5)$$

In terms of polarimetric feature selection, this study combines FP and compact-polarimetric SAR features to construct eight polarimetric feature channels for each single-temporal phase. The compact-polarimetric features include four compact-polarimetric backscattering coefficients under RH, RV, RR, and RL configurations, as well as the first component of the compact-polarimetric Stokes vector g.

The fusion of FP and CP features in this study is physically grounded. FP data (HH, HV, VH, and VV) operate in the linear polarization basis and are highly sensitive to geometric scattering structures, while CP features (RH, RV, RR, RL, and Stokes parameters) represent circular-polarization observations, with Stokes descriptors capturing depolarization effects linked to canopy complexity. Thus, FP and CP provide complementary information—geometric mechanisms versus statistical depolarization and helicity—which is particularly beneficial for distinguishing rice varieties with different structural and phenological traits.

B. Study Area and Data Sources

The study area is located at the junction of Xuyi, Hongze, and Jinhu counties in Huai'an, Jiangsu Province, China, within a typical rice-growing region of the Huaihe River Plain. The area is characterized by relatively flat terrain and concentrated paddy field distribution, together with complex background land-cover types such as water, shoals, and urban areas. These conditions make the area suitable for evaluating fine-grained rice classification under multi-temporal polarimetric SAR observations. In this study, the main rice planting types are categorized as D-J and T-H. These two rice types exhibit similar backscattering responses and field morphologies at certain growth stages, making them prone to confusion in pixelwise classification. Accordingly, the experimental scenario in this study is not intended merely to assess general land-cover segmentation, but is directly oriented toward fine-grained discrimination between D-J and T-H rice types.

This study uses six RADARSAT-2 C-band fully polarimetric SAR images acquired during the 2015 rice-growing season. All images were collected in the FQ20W imaging mode, with

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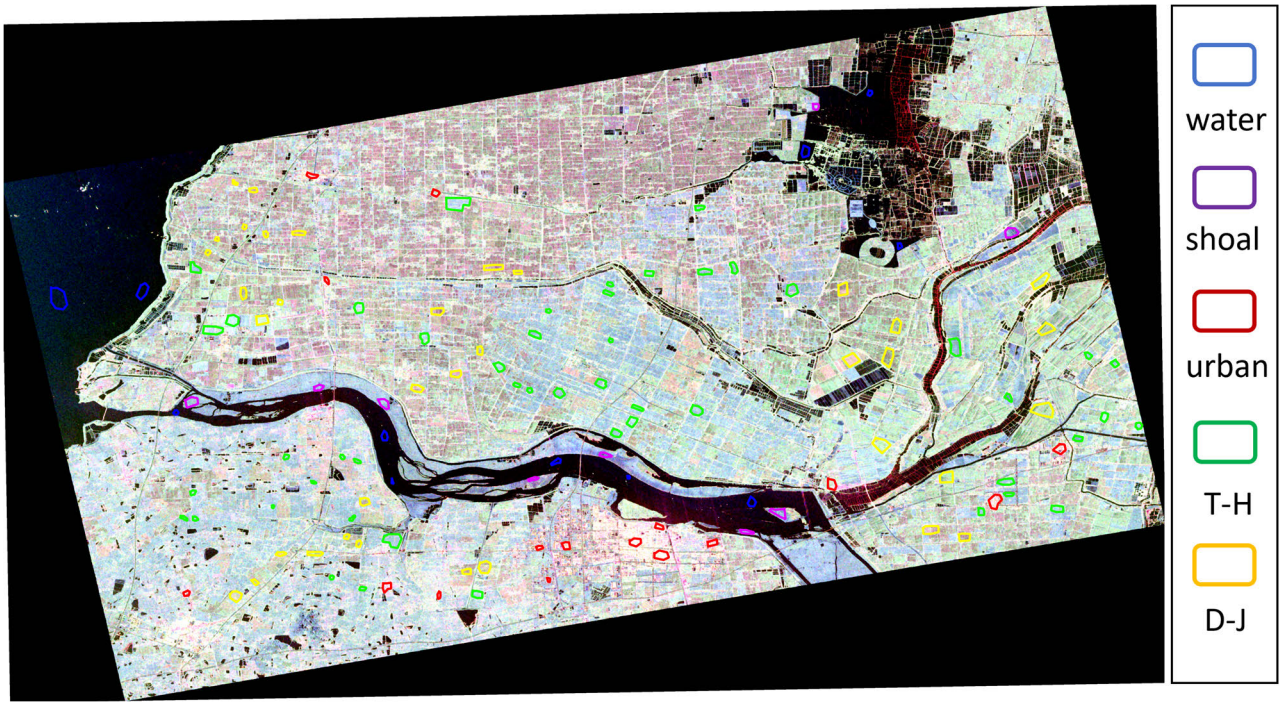


Fig. 1. Spatial distribution of the field-survey polygons over the RADARSAT-2 pseudocolor image of the study area acquired on June 12, 2015. The ground-reference polygons collected during the 2015 field survey are overlaid on the image. T-H, D-J, water, urban, and shoal samples are indicated by green, yellow, blue, red, and purple polygon boundaries, respectively.

TABLE I
MULTI-TEMPORAL FULL-POLARIZATION SAR DATA

Data Acquisition Time	Day of Year	Imaging Mode	Pixel Size(A×R, m)	Incidence Angle(°)
2015/06/12	163	FQ20W	5.2×7.6	38–41
2015/07/30	211	FQ20W	5.2×7.6	38–41
2015/08/23	235	FQ20W	5.2×7.6	38–41
2015/09/16	259	FQ20W	5.2×7.6	38–41
2015/10/10	283	FQ20W	5.2×7.6	38–41
2015/11/03	307	FQ20W	5.2×7.6	38–41

a pixel spacing of 5.2 × 7.6 m and an incidence angle range of 38°–41°. Information on the acquisition dates, imaging mode, pixel spacing, and incidence angles of the six SAR images is summarized in Table I. The six temporal phases cover the main phenological stages of rice, from the seedling and vegetative growth stages to maturity and harvest. This time-series dataset was selected to enable the model to jointly observe stage-dependent changes in rice canopy structure, field water conditions, and polarimetric scattering responses, thereby providing cross-temporal evidence for distinguishing D-J from T-H. The spatial distribution of the ground-reference polygons over the RADARSAT-2 pseudocolor image is shown in Fig. 1.

All multi-temporal SAR data were preprocessed following a standard polarimetric pipeline: 1) radiometric calibration using header-derived parameters; 2) extraction of the FP covariance matrix and CP Stokes vector from calibrated SLC data; 3) speckle suppression using a 5 × 5 refined Lee filter; and 4) terrain correction with SRTM DEM (WGS-84 and 10-m resolution), followed by second-order polynomial tie-point refinement for multi-temporal coregistration (RMSE <0.5 pixels). The CP parameters were simulated from FP data using the CTLR mode under the reciprocity assumption, and were calibrated and filtered independently to ensure consistent processing. As all features originate from the same coregistered SLC imagery, their pixelwise correspondence is inherently guaranteed.

From a phenological perspective, the six temporal phases cover the main growth stages from seedling development to maturity and harvest, allowing the model to capture scattering differences caused by temporal variations in canopy structure, water status, and field background. Although D-J and T-H may show similar backscattering intensities at individual temporal phases, their cross-temporal change rates, channel-energy distributions, and structural evolution are not fully identical. Therefore, joint modeling of multiple temporal

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phases is more suitable than single-temporal classification for revealing fine-grained differences between the two rice types [7], [8].

During the RADARSAT-2 overpasses, synchronous field surveys were conducted throughout the study area. The field investigation contains 611 georeferenced field-survey records, including 47 water, 40 shoal, 73 urban, 211 direct-sown japonica rice paddy (D-J), and 240 transplanting hybrid rice paddy (T-H) records. These field-survey records represent in situ observations and should be distinguished from the annotation polygons used for model training and evaluation.

The final annotation dataset used in this study contains 121 independent ground-reference polygons, including 36 D-J polygons, 49 T-H polygons, and 36 nonrice polygons corresponding to water, urban, and shoal categories. Their spatial distribution is shown in Fig. 1. As illustrated in the figure, the annotated polygons are distributed across different parts of the study area rather than being concentrated in a single local region.

The 121 annotation polygons cover a total labeled area of approximately 1.975 km^2. Because all geometrically corrected SAR data were resampled to a spatial resolution of 10×10 m, each labeled pixel represents approximately 100 m^2 on the ground. The annotated regions therefore correspond to approximately 19 750 labeled pixels. The mean area of an annotation polygon is approximately 1.63×10^4 m^2, corresponding to an equivalent square field of approximately 128×128 m, or approximately 163 labeled pixels per polygon on average. These statistics indicate that the annotation polygons represent field-scale spatial units rather than isolated point samples.

Considering the strong spatial autocorrelation among neighboring pixels within the same agricultural field, the dataset was partitioned at the polygon level rather than by randomly dividing individual pixels or image patches. The 121 annotation polygons were divided into 73 training polygons, 24 validation polygons, and 24 test polygons. All pixels, multitemporal observations, and image patches derived from the same polygon were assigned exclusively to one subset. Consequently, no pixels or patches from the same field polygon were shared among the training, validation, and test subsets.

The six RADARSAT-2 acquisitions provide repeated temporal observations of the same ground-reference polygons and are therefore not treated as additional independent field samples. Instead, the multi-temporal observations provide complementary temporal and polarimetric information for characterizing rice phenological evolution and distinguishing D-J from T-H.

III. Methodology

A. Overall Architecture of PC-MTPNet

To address class confusion, fragmented boundaries, and discontinuities in sliding-window inference for fine-grained rice classification from multi-temporal polarimetric SAR images, this study proposes PC-MTPNet, a prototype-constrained framework. PC-MTPNet is designed to address three coupled sources of error: insufficient explicit modeling of multitemporal polarimetric structure, the absence of prototype constraints in pixelwise losses, and spatial discontinuities caused by SAR speckle noise and patch stitching. Accordingly, while preserving the original 48-channel multi-temporal polarimetric SAR input, PC-MTPNet introduces MTP-Desc, MTP-Polar, boundary auxiliary supervision, and Hann–Gaussian sliding-window fusion to improve the statistical separability of rice subclasses and the stability of full-scene prediction.

As shown in Fig. 2, the input of PC-MTPNet consists of six observation phases, each with eight polarimetric features, forming a 48-channel SAR input tensor. Temporal information is incorporated through two complementary pathways. First, the six acquisitions are arranged in chronological order and stacked as a 48-channel input, allowing the segmentation backbone to learn cross-date feature combinations together with polarimetric and spatial context. Second, MTP-Desc explicitly restores the phase–channel structure of the input and calculates phasewise statistics, temporal fluctuations, and adjacent-phase variations. MTP-Polar subsequently performs prototype-constrained learning in this multi-temporal statistical descriptor space. The detail-refined U-Net++ backbone takes the original 48-channel data as input and produces six-class logits and probability maps for background, water, shoal, urban, D-J, and T-H. In parallel, MTP-Desc constructs an 8-D MTP-Desc from the same input to characterize scattering intensity, channel dispersion, channel entropy, and adjacent-phase variation. MTP-Polar further computes class prototypes in this statistical descriptor space and enhances intraclass compactness and interclass separability through prototype alignment and a D-J/T-H margin constraint. In the overall workflow, PC-MTPNet first performs six-class semantic segmentation using the detail-refined U-Net++ backbone. It then strengthens the multi-temporal polarimetric statistical differences between D-J and T-H through MTP-Desc and MTP-Polar. The edge head and boundary loss are used to improve segmentation quality along rice field boundaries, shoal transition zones, and urban edge regions. Finally, Hann–Gaussian weighted sliding-window fusion is applied during full-scene inference to reduce patch stitching artifacts and generate continuous and stable scene-level probability maps. Thus, the backbone preserves high-dimensional SAR information from the original 48-channel input, whereas the MTP-Desc statistical space provides a stable and interpretable metric basis for MTP-Polar.

B. Multi-Temporal Polarimetric Input and Statistical Description

Temporal information is used through two complementary mechanisms. First, the six SAR acquisitions are arranged in chronological order from T1 to T6, with eight polarimetric features retained for each acquisition. The resulting 48-channel tensor is directly fed into the detail-refined U-Net++ backbone. Because each 2-D convolution jointly operates on all input channels, the backbone can implicitly learn cross-date combinations together with spatial and polarimetric features. This pathway can be regarded as fixed-order multidate feature fusion rather than recurrent or 3-D temporal modeling.

Second, MTP-Desc explicitly operates along the temporal dimension. The 48-channel tensor is reorganized into six temporal phases and eight polarimetric channels per phase.

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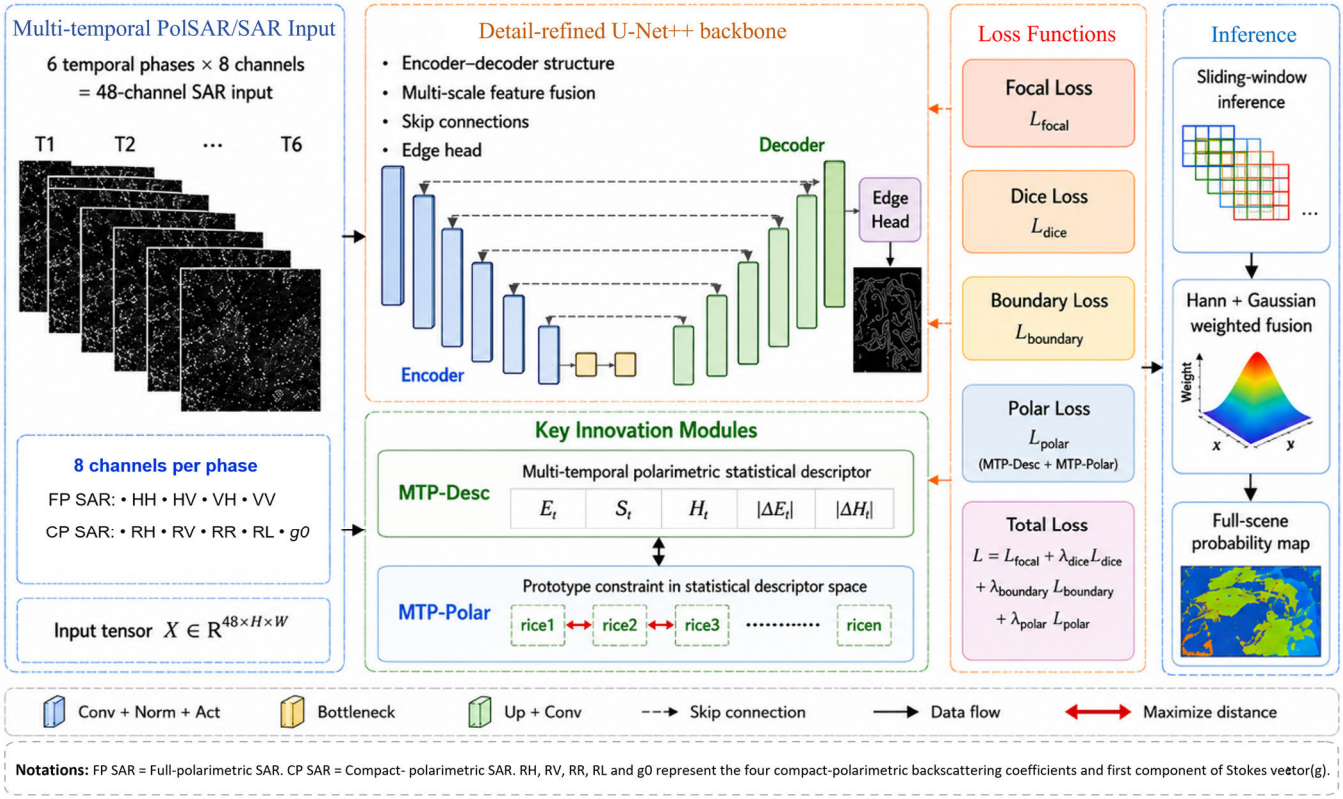


Fig. 2. Overall architecture of PC-MTPNet for multi-temporal PolSAR rice fine-grained classification.

Phasewise energy E_t, channel dispersion S_t, and channel entropy H_t are first calculated for each acquisition. Their temporal averages and fluctuations, together with the adjacent-phase variations $|\Delta E_t|$ and $|\Delta H_t|$, are then summarized to construct the 8-D MTP-Desc vector. In this manner, MTP-Desc explicitly represents both the overall temporal state and the changes between consecutive rice growth phases.

The resulting descriptor is used by MTP-Polar during training to construct class prototypes and impose prototype alignment and D-J/T-H margin separation. MTP-Desc and MTP-Polar therefore provide a training-time statistical constraint and do not replace the original 48-channel input. During inference, the chronologically ordered 48-channel data are processed by the trained segmentation backbone, and Hann–Gaussian fusion is used only to integrate spatially overlapping prediction patches.

The input of PC-MTPNet consists of six observation phases, each containing eight full- and compact-polarimetric features. Rather than treating the 48 channels as a conventional multichannel image, this study explicitly restores their phase–channel structural organization. This formulation allows the model to jointly capture the temporal evolution of scattering signatures throughout the rice-growing season as well as polarimetric response discrepancies within individual phases. Formally, the input tensor is defined as

$$X \in \mathrm{R}^{B\times T\times K\times H\times W} \tag{6}$$

where B denotes the batch size, H and W represent the image height and width, T is the number of observation phases, and K stands for the number of polarimetric statistical channels per phase. The original 48-channel SAR tensor continues to be fed into the detail-refined U-Net++ backbone. As introduced above, MTP-Desc builds a low-dimensional statistical space exclusively during training to enable the subsequent prototype constraint, instead of substituting the original input. MTP-Desc is a MTP-Desc designed for fine-grained rice classification. It focuses on four types of statistics directly relevant to this task: overall scattering intensity, channel response dispersion, channel-energy distribution complexity, and adjacent-phase variation. For pixel p, let $x_{t,k}^2(p)$ denote the input value of the Kth channel at the tth phase. This study defines

$$\begin{aligned} E_t(p) &= \log\left(1+\frac{1}{K}\sum_{k=1}^{K}x_{t,k}^2(p)\right) \\ S_t(p) &= \sqrt{\frac{1}{K}\sum\nolimits_{k=1}^{K}\left(x_{t,k}(p)-m_t(p)\right)^2} \\ H_t(p) &= -\frac{1}{\log K}\sum_{k=1}^{K}q_{t,k}(p)\log\left(q_{t,k}(p)+\varepsilon\right) \end{aligned} \tag{7}$$

where $E_t(p)$ describes the overall scattering intensity at the tth phase, $S_t(p)$ describes the response dispersion among different channels within the same phase, and $H_t(p)$ describes the complexity of channel-energy distribution. Log energy is used to compress extreme responses caused by local strong scattering and speckle noise, while channel dispersion and channel entropy characterize structural differences among polarimetric/statistical channels.

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Finally, the 8-D MTP-Desc descriptor of pixel p is defined as

$$D(p) = \begin{bmatrix} \text{mean}_t(E_t), \text{std}_t(E_t) \\ \text{mean}_t(S_t), \text{std}_t(S_t) \\ \text{mean}_t(H_t), \text{std}_t(H_t) \\ \text{mean}_t(|\Delta E_t|), \text{mean}_t(|\Delta H_t|) \end{bmatrix}. \quad (8)$$

Equation (8) forms the core statistical space of MTP-Polar, where ΔE and ΔH denote the adjacent-phase variations of the polarimetric energy descriptor and the polarimetric entropy descriptor, respectively. They are used to characterize the temporal changes in polarimetric statistical properties between two adjacent growth phases. It compresses multi-temporal polarimetric SAR information into four types of evidence, namely average state, temporal fluctuation, channel structure, and dynamic variation. As a result, D-J/T-H differences no longer rely entirely on implicit learning from the 48 input channels, but are explicitly represented in a physically meaningful descriptor space.

To provide an intuitive interpretation of these temporal statistics, the phasewise profiles of E_t, S_t, and H_t, together with the adjacent-phase variations $|\Delta E_t|$ and $|\Delta H_t|$, are further visualized and analyzed in Section IV-B. The visualization is based on polygon-level aggregation so that spatially correlated pixels within the same field are not treated as independent observations.

C. Prototype Constraint and Backbone Network

In the MTP-Desc statistical space, this study further constructs the MTP-Polar prototype constraint. MTP-Polar constrains class distributions using ground-truth and prediction-derived prototypes, while imposing task-specific margin separation between D-J and T-H. The basic idea of prototype learning and margin-based constraints follows related classical studies [30], [31], [32], and is adapted here to rice-subclass regularization in the multi-temporal polarimetric descriptor space.

Before formulating MTP-Polar, we further clarify the distinction between the proposed MTP-Desc/MTP-Polar design and existing descriptors or prototype-based losses. Although MTP-Desc is constructed from multi-temporal polarimetric statistics, it is different from conventional polarimetric feature descriptors. Traditional descriptors are usually used as independent handcrafted input features for classifiers, whereas MTP-Desc organizes the original polarimetric SAR observations from six temporal phases and eight channels into an explicit statistical description space for deep constraint learning. It jointly considers temporal evolution, channel dispersion, energy distribution complexity, entropy-related information, and adjacent-phase variation, and provides an interpretable basis for subsequent prototype-constrained optimization.

MTP-Polar is also different from conventional prototype-based losses, center loss, and general metric-learning losses. Existing prototype-based constraints usually operate on generic deep features and mainly encourage samples to approach class centers. On the contrary, MTP-Polar performs prototype alignment in the MTP-Desc statistical descriptor space, where the prototypes are associated with multi-temporal polarimetric characteristics of rice subclasses. In addition, MTP-Polar introduces margin separation between D-J and T-H rice subclasses, which is specifically designed for fine-grained rice-type discrimination rather than general semantic segmentation.

Let $y_c(p)$ denote the ground-truth mask of class c, and $p_c(p)$ denote its predicted probability. The ground-truth prototype and prediction-derived prototype of class c in the MTP-Desc space are defined as

$$\mu_c^{gt} = \frac{\sum_p Y_c(p) D(p)}{\sum_p Y_c(p) + \varepsilon}$$
$$\mu_c^{pred} = \frac{\sum_p P_c(p) D(p)}{\sum_p P_c(p) + \varepsilon} \quad (9)$$

where μ_c^{gt} represents the class center of the ground-truth samples in the statistical descriptor space and μ_c^{pred} represents the class center weighted by the predicted probabilities. To jointly achieve class alignment and rice-subclass separation, the MTP-Polar loss is formulated as

$$L_{polar} = \sum_{c \in C} \text{SmoothL1}\left(\mu_c^{pred}, \mu_c^{gt}\right) + \lambda_m \max\left(0, m - \left\|\mu_{D-J}^{pred} - \mu_{T-H}^{pred}\right\|_2\right) \quad (10)$$

where SmoothL1(·) denotes the smooth L1 loss. For an error term x, it is defined as SmoothL1(x) = $0.5x^2$ if |x| < 1, and SmoothL1(x) = |x| − 0.5 otherwise. This loss behaves as an L2 loss for small errors and as an L1 loss for large errors, which reduces the sensitivity to outliers while maintaining stable gradients. Then, the first term aligns the predicted class distribution with the ground-truth class distribution, and the second term imposes a margin constraint only on the two rice classes, D-J and T-H.

When the prototype distance between the two rice classes is smaller than m, an additional penalty is applied, encouraging clearer statistical separation in the MTP-Desc space. Compared with ordinary pixelwise losses, MTP-Polar directly constrains the overall class distributions, making it more suitable for addressing scattering similarity, interlaced boundaries, and probabilistic confusion between rice subclasses.

Fig. 3 illustrates the mechanism of MTP-Polar. Class prototypes are first constructed in the MTP-Desc space. Prototype alignment is then used to improve intraclass consistency, while D-J/T-H margin separation reduces mutual confusion between the two rice types.

PC-MTPNet adopts a detail-refined U-Net++ backbone based on the nested encoder–decoder architecture of U-Net++ [39]. The input layer is adapted to RV the original 48-channel multi-temporal polarimetric SAR tensor, which consists of six observation phases and eight polarimetric features per phase. An auxiliary edge head is attached to the decoder to strengthen the representation of field boundaries, while boundary loss is applied during training as auxiliary boundary supervision. The backbone produces six-class logits and probability maps for background, water, shoal, urban, D-J, and T-H. In this manuscript, the term "detail-refined" describes this task-specific adaptation of U-Net++ and is not intended as the proper name of an independently published network.

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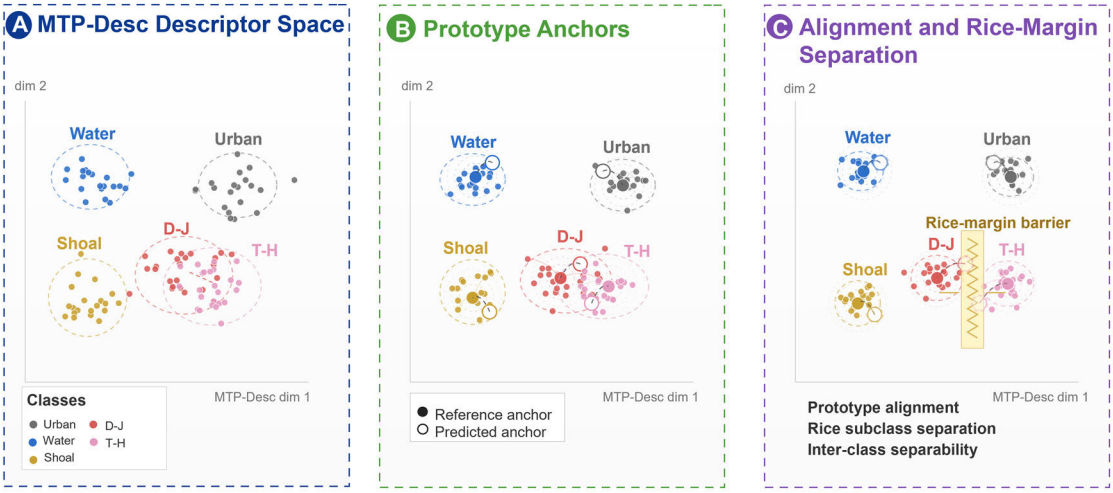


Fig. 3. MTP-Polar prototype constraint in the MTP-Desc statistical descriptor space.

MTP-Desc and MTP-Polar operate in parallel to further enhance the intraclass consistency and interclass separation of the rice subclasses in the statistical descriptor space.

Considering that speckle noise, mixed pixels, and scattering from adjacent land-cover types in SAR imagery can weaken field boundary stability, this study retains the edge head and boundary loss as auxiliary boundary supervision [26], [33], [34]. The edge head produces boundary responses along class interfaces, while boundary loss enforces consistency between predicted and annotated boundaries. This reduces class leakage and patch fragmentation along water edges, shoal transition zones, urban boundaries, and D-J/T-H interfaces.

D. Loss Function and Sliding-Window Inference Strategy

To jointly optimize class discrimination, regional overlap, boundary quality, and statistical prototype distributions, PC-MTPNet adopts a composite loss function and applies Hann–Gaussian weighted sliding-window fusion during inference. This section presents the composite loss terms and the sliding-window inference strategy together. In particular, focal loss, dice loss, boundary loss, and polar loss [34], [35], [36] are combined to improve label segmentation performance, with particular emphasis on enhancing the separability of different rice subclasses. The composite loss is expressed as

$$L = L_{focal} + \lambda_{dice} L_{dice} + \lambda_{boundary} L_{boundary} + \lambda_{polar} L_{polar} \quad (11)$$

where L_{focal} mitigates class imbalance and reduces the dominance of easily classified samples during training, encouraging the model to focus more on difficult pixels; L_{dice} improves the overlap between predicted and annotated regions; $L_{boundary}$ enhances the consistency between predicted and annotated boundaries; and L_{polar} constrains class prototype distributions in the MTP-Desc statistical descriptor space.

Each loss term targets a specific source of error in fine-grained rice classification from SAR imagery. Focal loss targets class imbalance and difficult pixels, Dice loss focuses on patch completeness, boundary loss addresses field boundaries and transition zones, and polar loss reduces statistical confusion between rice subclasses. Through this division of roles, PC-MTPNet connects the pixel–region–boundary hierarchy in conventional segmentation optimization with the class-prototype hierarchy in the multi-temporal polarimetric statistical space.

Because the SAR images of the study area cover a large spatial extent, directly feeding the full scene into the network is constrained by GPU memory. Therefore, sliding-window inference is adopted to obtain full-scene prediction results. Conventional sliding-window inference usually stitches or averages the predictions of different patches, which can easily introduce block artifacts between adjacent patches. This issue is more pronounced in SAR imagery because SAR data contain inherent speckle noise, and patch-edge regions may exhibit probability fluctuations due to insufficient contextual information.

To reduce discontinuities caused by sliding-window stitching, PC-MTPNet adopts a Hann–Gaussian weighted sliding-window fusion [27], [37]. This strategy is applied only during inference and does not alter the training labels, class system, or network parameters. For large-scale multi-temporal polarimetric SAR scenes, Hann–Gaussian weighted sliding-window fusion effectively suppresses patch stitching artifacts and produces more continuous and stable probability maps. In particular, along rice-subclass boundaries and in fragmented paddy field regions, this strategy improves the spatial consistency of probability maps, providing more reliable full-scene outputs for subsequent confusion matrix analysis, probability map analysis, and visualization of fine-grained rice classification results.

As shown in Fig. 4, Hann–Gaussian weighted sliding-window inference is used for full-scene prediction. First, the full multi-temporal SAR image is divided into mutually overlapping inference patches. Second, a confidence weight map with higher weights at the patch center and lower weights near the patch boundaries is constructed within each patch, so that central predictions contribute more strongly during fusion, while potentially unstable predictions near patch edges are naturally suppressed. Finally, class probabilities from all

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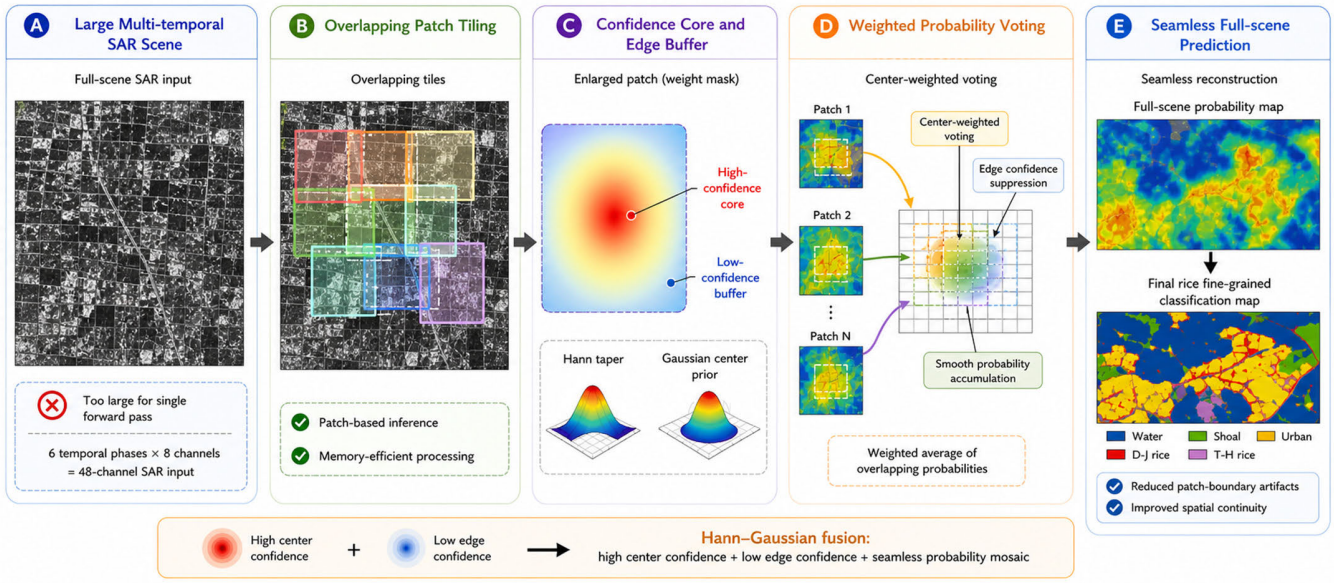


Fig. 4. Hann–Gaussian weighted sliding-window inference for full-scene prediction.

overlapping patches are integrated through weighted voting to generate a spatially continuous full-scene probability map and the final classification map. This strategy is applied only during inference and does not modify the training labels, class system, or network parameters. Nevertheless, it effectively improves the stability of full-scene prediction along rice field boundaries and fragmented paddy regions.

Based on the methods proposed in Section III, the subsequent experiments and result analyses are designed to correspond directly to each component of PC-MTPNet. The overall metrics are used to evaluate the six-class segmentation capability of the complete framework. The D-J/T-H classification accuracies are used to assess the fine-grained separation capability of MTP-Desc and MTP-Polar. Probability heatmaps are used to examine whether the statistical constraint improves the class probability fields. The confusion matrix is used to analyze whether residual errors remain concentrated between rice subclasses. Boundary and postprocessing results are further used to evaluate the contributions of the edge head, boundary loss, and sliding-window fusion to spatial continuity.

IV. Experimental and Results Analysis

A. Experimental Settings and Evaluation Metrics

To validate the effectiveness of PC-MTPNet for fine-grained rice classification from multi-temporal polarimetric SAR images, the experiments are organized around five aspects: overall segmentation capability, rice-subclass separability, spatial probability structure, sources of residual errors, and boundary stability. The compared methods include CVNN-PolSAR [38], U-Net++ based on a nested U-Net architecture [39], RandomForest [15], SegFormer-B1, and SegFormer-B0 [21]. PC-MTPNet is the proposed method. U-Net++ is used to quantify the gains introduced by MTP-Desc and MTP-Polar. CVNN-PolSAR serves as a baseline for polarimetric-compatible modeling. SegFormer-B1 and SegFormer-B0 represent strong and lightweight Transformer-based segmentation baselines, respectively. RandomForest is used as a traditional machine learning reference.

To ensure the reproducibility of the experiments, the main implementation settings of PC-MTPNet are summarized as follows. The input of PC-MTPNet consists of 48 channels, corresponding to six temporal phases and eight polarimetric features for each phase. The input patch size is set to 384×384. The model is optimized using the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. The batch size is set to 2, and a cosine annealing learning-rate scheduler is adopted during training. The random seed is fixed to 42 for the reported main experiment. The rice refinement model is trained for 120 epochs. The final full-scene prediction is generated using the weighted sliding-window fusion strategy.

The dataset partition was performed using the complete annotation polygon as the independent spatial unit. Among the 121 annotation polygons, 73 were assigned to the training subset, 24 to the validation subset, and 24 to the independent test subset. The training subset was used for network parameter optimization, the validation subset was used for model selection and convergence assessment, and the test subset was used only for the final quantitative evaluation. All pixels, image patches, and six-temporal-phase observations associated with the same annotation polygon were retained in the same subset. Therefore, spatially correlated pixels or overlapping patches derived from one polygon could not simultaneously appear in the training, validation, and test subsets. All comparison methods and ablation variants were trained and evaluated using the identical polygon-level partition to ensure a fair comparison.

The evaluation metrics consist of two parts: overall classification metrics and rice-specific classification metrics. The overall metrics include overall accuracy (OA), mean intersection over union (mIoU), macro-averaged $F1$ score (Macro-F1),

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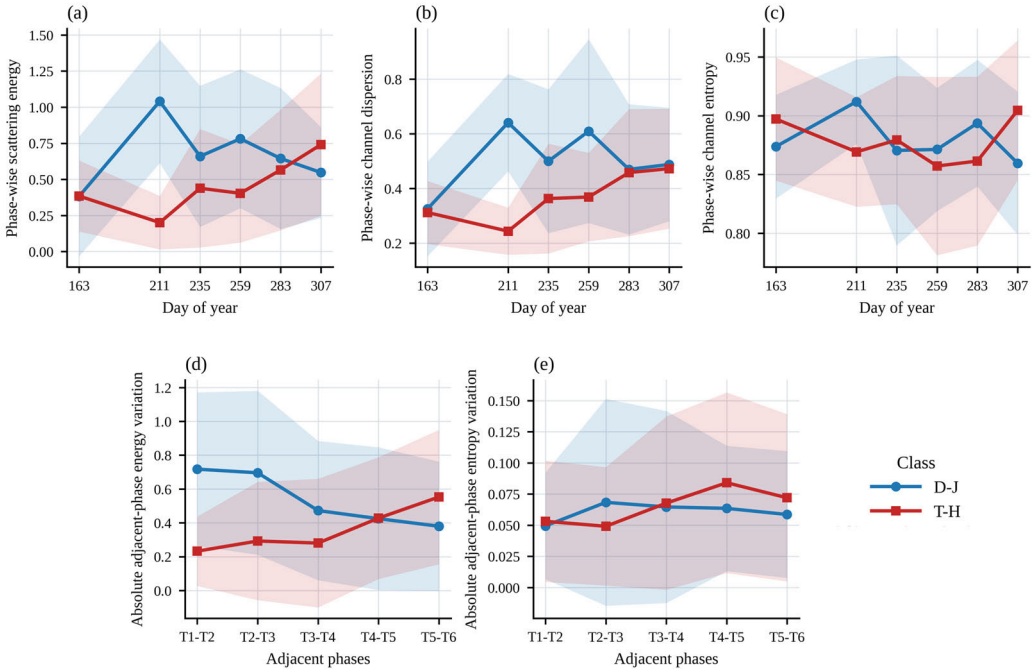


Fig. 5. Temporal evolution of the MTP-Desc statistical parameters for direct-sown japonica rice paddy (D-J) and transplanting hybrid rice paddy (T-H). (a) E_t. (b) S_t. (c) H_t. (d) $|\Delta E_t|$. (e) $|\Delta H_t|$.

and Cohen's kappa coefficient (Kappa). OA reflects the overall correctness of valid pixels; mIoU emphasizes multiclass region-overlap quality; Macro-F1 measures class-balanced performance; and Kappa evaluates the agreement between predictions and reference labels after accounting for chance agreement. In addition, the reported rice-specific metrics include D-J accuracy, T-H accuracy, D-J/T-H mean accuracy, and D-J/T-H minimum accuracy. The first two metrics measure the classwise recognition capability for the two rice types, while the latter two are used to assess whether both rice types are recognized stably, preventing high accuracy for one rice type from masking performance degradation for the other.

Furthermore, the image-based classification results are used to provide spatial interpretations of the quantitative metrics. Classwise accuracy results enable simultaneous inspection of nonrice classes, rice subclasses, and overall performance. Full-scene prediction rendering is used to assess whether large-scale land-cover patches remain spatially coherent. Probability heatmaps are used to analyze whether the probability fields of D-J and T-H exhibit field-level complementarity. The class accuracy matrix is used to identify the recognition level of each category. Auxiliary postprocessing maps are used to discuss the contribution of boundary optimization to SAR segmentation stability.

B. Temporal Evolution Analysis of MTP-Desc Parameters

To provide a clearer interpretation of the temporal information represented by MTP-Desc, Fig. 5 visualizes the phasewise polarimetric statistical profiles of D-J and T-H over the six RADARSAT-2 acquisition dates. The visualization includes the scattering-energy descriptor E_t, channel-dispersion descriptor S_t, channel-entropy descriptor H_t, and the adjacent-phase variations $|\Delta E_t|$ and $|\Delta H_t|$.

The statistics were calculated at the polygon level. In particular, the pixelwise descriptor values were first averaged within each reference polygon at each temporal phase. The classwise mean and standard deviation were then calculated across the D-J and T-H polygons, respectively. Therefore, pixels within the same field polygon were not treated as independent observations. The solid lines in Fig. 5(a)–(c) represents the classwise means over the six acquisition dates, while the shaded regions indicate the standard deviations among the reference polygons. Fig. 5(d) and (e) presents the corresponding absolute changes between two adjacent acquisition phases.

As shown in Fig. 5, D-J and T-H may exhibit partially overlapping statistical responses at individual acquisition dates, but their temporal trajectories and adjacent-phase variations are not identical throughout the growing season. The differences are reflected not only in the average scattering-energy level but also in the temporal fluctuation of channel dispersion, channel-energy entropy, and the magnitude of adjacent-phase changes. These observations support the design of MTP-Desc, which jointly encodes the average temporal state, temporal fluctuation, channel structure, and adjacent-phase dynamic variation rather than relying on the response of a single acquisition date.

The visualized phasewise profiles are the intermediate temporal statistics used to construct the final 8-D descriptor in (8). The eight descriptor dimensions are temporal summary statistics rather than eight temporal observations. Therefore,

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TABLE II
COMPARISON OF QUANTITATIVE CLASSIFICATION ACCURACY METRICS FOR DIFFERENT METHODS (%)

Method	OA	mIoU	Macro-F1	Kappa
PC-MTPNet (Ours)	**93.73**	**86.49**	**92.66**	**92.11**
CVNN-PolSAR	90.68	81.38	89.86	91.56
SegFormer-B1	89.71	79.16	88.16	87.07
U-Net++	81.93	67.09	79.83	77.39
RandomForest	80.92	74.56	69.63	68.66
SegFormer-B0	70.15	51.88	67.34	62.79

TABLE III
ALL CATEGORIES ACCURACY (%) COMPARISON OF DIFFERENT METHODS

Method	Water	Urban	Shoal	D-J	T-H	OA	mIoU
PC-MTPNet (Ours)	**96.58**	**93.10**	**94.30**	**93.50**	**92.49**	**93.73**	**86.49**
CVNN-PolSAR	93.63	92.62	90.42	90.85	89.28	90.68	81.38
SegFormer-B1	93.92	87.99	91.45	87.57	85.12	89.71	79.16
U-Net++	97.95	82.40	84.48	75.93	75.99	81.93	67.09
RandomForest	96.77	69.63	80.92	68.66	67.44	80.92	74.56
SegFormer-B0	81.68	64.55	71.53	62.95	67.16	70.15	51.88

TABLE IV
SPECIFIC IDENTIFICATION RESULTS OF TWO RICE TYPES (D-J AND T-H) (%)

Method	D-J/T-H mean	D-J/T-H min
PC-MTPNet (Ours)	**93.00**	**92.49**
CVNN-PolSAR	90.07	89.28
SegFormer-B1	86.35	85.12
U-Net++	75.96	75.93
RandomForest	68.05	67.44
SegFormer-B0	65.06	62.95

visualizing E_t, S_t, H_t, $|\Delta E_t|$, and $|\Delta H_t|$ over the acquisition sequence provides a more direct interpretation of how temporal information is represented by MTP-Desc.

C. Overall Performance of Fine-Grained Rice Classification

Tables II–IV jointly report the complete quantitative values used to evaluate the compared methods. In particular, Table II reports the overall performance metrics, including OA, mIoU, Macro-F1, and Kappa; Table III reports the classwise accuracies of water, urban, shoal, D-J, and T-H; and Table IV reports D-J/T-H mean accuracy and D-J/T-H minimum accuracy. In Table III, classwise accuracy is calculated as the number of correctly classified pixels of a given class divided by the total number of reference pixels of that class, corresponding to producer accuracy/recall, i.e., the diagonal accuracy of the confusion matrix, rather than user accuracy. These metrics are used together to evaluate whether a method achieves high overall performance while maintaining stable recognition of both rice subclasses.

Table II illustrates the comparison of the overall classification performance of the six methods under the same reference-map setting, while Tables III and IV further provide classwise and rice-subclass stability results. PC-MTPNet achieves the best and most stable results, with an OA, mIoU, Macro-F1, and Kappa of 93.73%, 86.49%, 92.66%, and 92.11%, respectively. Compared with CVNN-PolSAR, PC-MTPNet improves OA and mIoU by 3.05 and 5.11 percentage points, indicating that polarimetric-compatible modeling alone is insufficient to fully address the fine-grained confusion between D-J and T-H. Its advantage over U-Net++ further highlights the effectiveness of the proposed multi-temporal polarimetric statistical description, prototype constraint, and boundary enhancement mechanisms.

The improvement of PC-MTPNet is not limited to a single rice subclass. Instead, it maintains consistently high performance for both D-J and T-H, which is consistent with the objective of MTP-Polar to enlarge the relative distance between the two rice types in the statistical descriptor space. On the contrary, RandomForest and SegFormer-B0 show lower performance, suggesting that models without explicit multi-temporal polarimetric structural constraints struggle to capture fine-grained intrarice differences.

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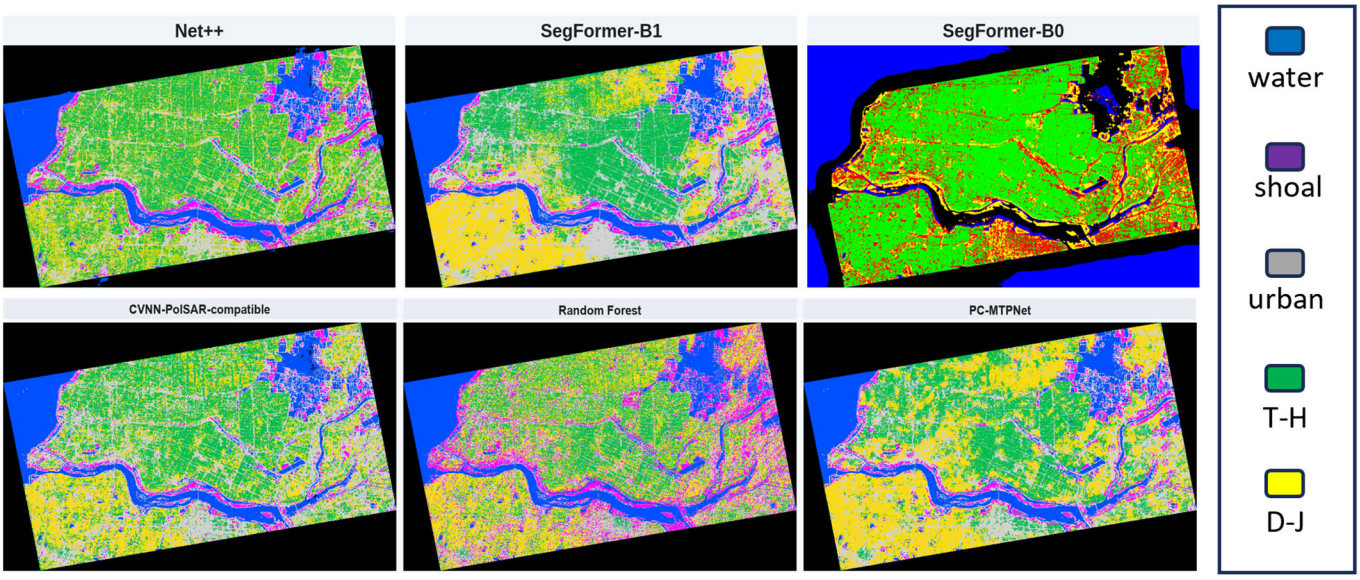


Fig. 6. Classification results of representative methods.

As presented in Table III, PC-MTPNet achieves high accuracies for D-J and T-H while retaining strong recognition performance for water, shoal, and urban. This indicates that the improvement comes from a more discriminative overall feature space, rather than from gains in a single category. Together, these results confirm that PC-MTPNet improves both overall segmentation and rice-subclass discrimination.

D. Classwise Accuracy Stability and Spatial Distribution Analysis

In terms of classwise accuracy, PC-MTPNet achieves 96.58%, 93.10%, and 94.30% for water, urban, and shoal, respectively. This indicates that the proposed method improves rice-subclass recognition without compromising the stability of nonrice land-cover classification. CVNN-PolSAR achieves corresponding accuracies of 93.63%, 92.62%, and 90.42%, respectively, also demonstrating the capability of polarimetric features to characterize scattering differences among water, urban, and shoal areas. PC-MTPNet further maintains advantages in these categories, suggesting that MTP-Desc and MTP-Polar do not introduce a local bias toward D-J/T-H alone. Instead, they improve the overall six-class distribution structure through a more effective multi-temporal polarimetric statistical space.

Fig. 6 presents the classification results of the compared methods. Compared with quantitative accuracy metrics, full-scene classification maps more directly reveal spatial coherence issues in fine-grained rice mapping from remote sensing images. Numerous isolated noisy patches within large water or urban regions indicate unstable local probability estimates. Similarly, excessive fragmentation within farmland areas can reduce the practical usability of the resulting rice type map. Fig. 7 further explains the spatial behavior of PC-MTPNet from the probability perspective. For water, shoal, and urban, high-probability responses should be spatially consistent with the corresponding land-cover morphology. For D-J and T-H, high-probability regions should exhibit field-scale differences rather than complete spatial overlap. These patterns show that MTP-Desc and MTP-Polar improve not only the final labels but also the organization of class probability distributions.

Table IV evaluates the balanced recognition performance for D-J and T-H. PC-MTPNet achieves a D-J/T-H mean accuracy of 93.00% and a minimum accuracy of 92.49%, indicating stable recognition of both rice subclasses. In comparison, the mean/min accuracies are 90.07%/89.28% for CVNN-PolSAR, 86.35%/85.12% for SegFormer-B1, 75.96%/75.93% for U-Net++, 68.05%/67.44% for Random-Forest, and 65.06%/62.95% for SegFormer-B0. These results show that conventional segmentation or machine learning methods alone are insufficient for reliably distinguishing rice subclasses with similar scattering responses and field morphologies.

All compared methods are evaluated using the same fixed spatial sampling strategy and evaluation protocol to ensure a fair comparison. Given the limited number of field plots and strong spatial correlation between adjacent rice parcels, all compared methods use the identical polygon-level train/validation/test partition described in Section IV-A. In addition, we report the mean accuracy and minimum accuracy for D-J and T-H as stability metrics for rice subcategories, preventing high accuracy on one rice type from concealing performance drops on the other.

From the perspective of the model mechanism, this result is consistent with the design rationale in Section III. The differences between D-J and T-H are not always reflected in single-temporal intensity contrasts, but are often encoded in cross-temporal change rates, channel-energy distributions, and scattering-structure evolution. MTP-Desc explicitly represents these latent differences through temporal means, temporal variances, channel dispersion, channel entropy, and adjacent-phase variation, while MTP-Polar further constrains the class

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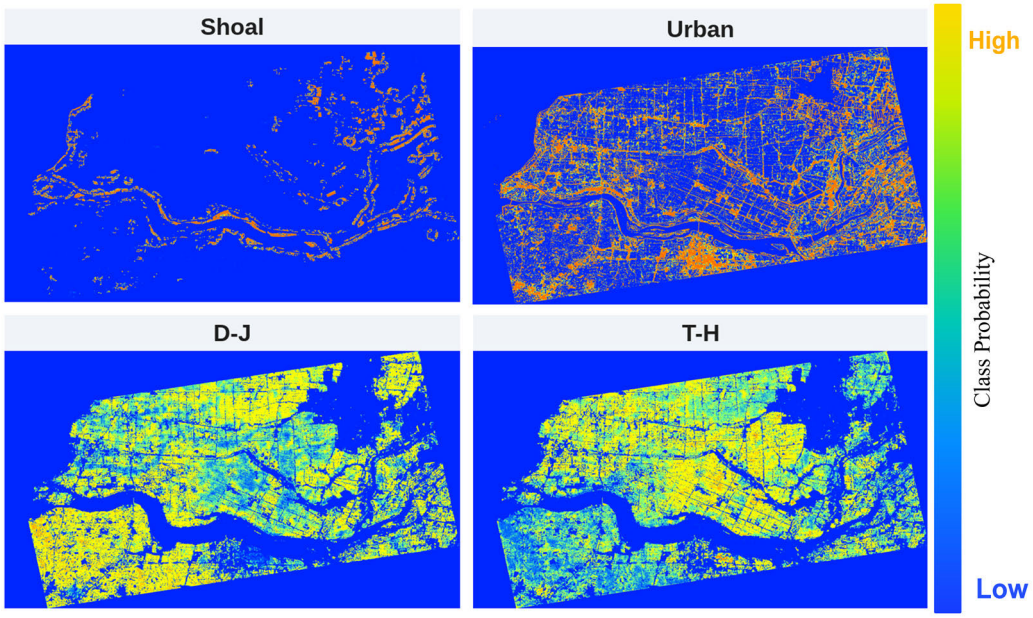


Fig. 7. Representative probability heatmaps of PC-MTPNet.

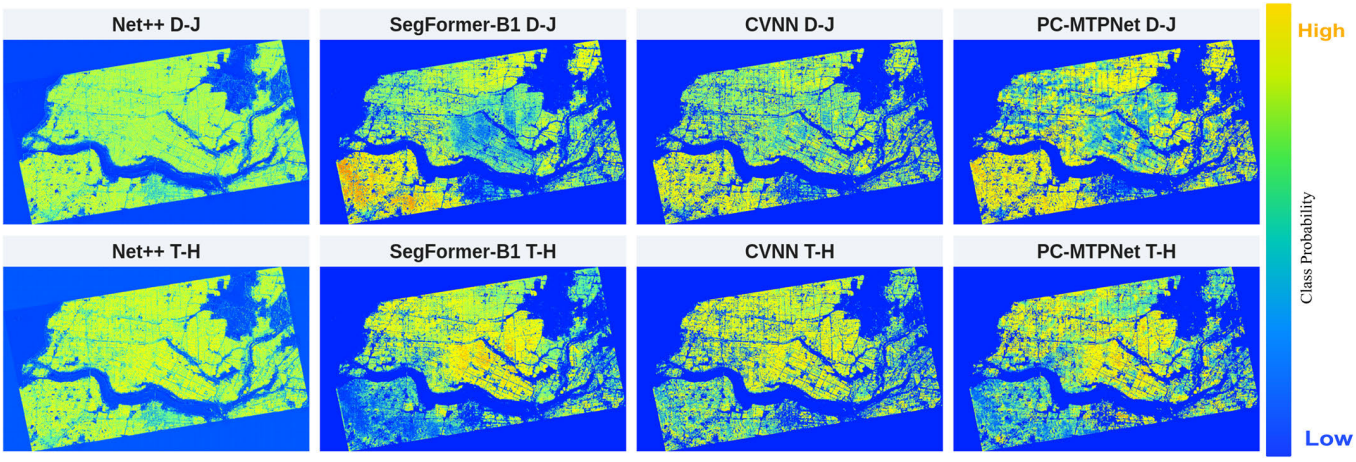


Fig. 8. Probability heatmap comparison of D-J and T-H rice subclasses across different methods.

prototype distance between D-J and T-H in the descriptor space. The probability heatmaps in Fig. 8 provides information that cannot be directly revealed by hard classification maps. If the D-J and T-H probability distributions produced by the same method are highly similar, the model may identify rice areas but still fail to separate rice subclasses reliably. On the contrary, if the two probability fields exhibit complementary patterns at the field scale, the model can form a clearer type-level separation.

E. Spatial Probability, Confusion Structure, and Boundary Stability Analysis

The confusion structure is used to interpret the sources of residual errors behind the high-accuracy results. For PC-MTPNet, the diagonal accuracies of D-J and T-H are substantially improved, yet mutual confusion between the two rice types remains an important component of the residual errors. This indicates that the fine-grained differences between D-J and T-H are not completely eliminated by the deep model itself, but are markedly alleviated through MTP-Desc and MTP-Polar. Fig. 9 presents the class accuracy matrix of PC-MTPNet under the identical reference protocol, providing a categorywise view of the recognition performance. The high diagonal accuracies for both nonrice classes and rice subclasses indicate that PC-MTPNet improves rice-subclass recognition while preserving reliable discrimination of background land-cover classes.

Auxiliary experiments were conducted to examine the effects of boundary supervision and inference-stage probability fusion on SAR segmentation stability. In SAR imagery, shoal areas, urban edges, and rice field boundaries are often affected by speckle noise, mixed pixels, and local scattering anomalies, which can lead to jagged boundaries, local patch noise, or spatial discontinuities in hard classification maps. To address

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TABLE V
ABSOLUTE PERFORMANCE OF ABLATION VARIANTS (%)

Variant	MTP-Desc	MTP-Polar	Edge Head	Boundary Loss	Hann–Gaussian Fusion	OA	mIoU	Macro-F1	Kappa	D-J Acc	T-H Acc	D-J/T-H Mean	D-J/T-H Min	ΔmIoU vs. A0
A0	✓	✓	✓	✓	✓	93.73	86.49	92.66	92.11	93.50	92.49	93.00	92.49	0.00
A1			✓	✓	✓	91.12	82.69	89.79	88.84	80.04	79.25	79.65	79.25	-3.80
A2	✓		✓	✓	✓	88.05	79.40	87.24	85.00	77.10	75.04	76.07	75.04	-7.09
A3	✓	✓		✓	✓	90.64	82.15	89.36	88.25	85.05	85.50	85.28	85.05	-4.34
A4	✓	✓	✓		✓	90.33	81.75	89.06	87.87	82.52	81.16	81.84	81.16	-4.74
A5	✓	✓			✓	90.45	82.17	89.40	88.01	78.97	76.76	77.87	76.76	-4.32
A6	✓	✓	✓	✓		93.74	86.51	92.67	92.12	93.26	92.23	92.75	92.23	+0.02

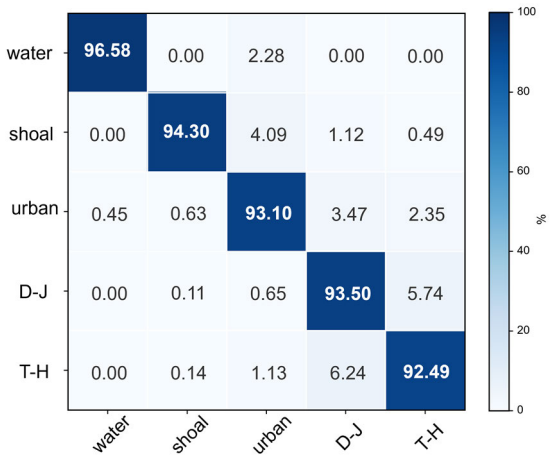


Fig. 9. Class accuracy matrix of PC-MTPNet under the identical reference protocol.

this issue, PC-MTPNet imposes constraints at both the training and inference stages. During training, the edge head and boundary loss strengthen boundary supervision along class interfaces. During inference, Hann–Gaussian sliding-window fusion reduces abrupt probability changes near patch edges, thereby improving spatial continuity of full-scene predictions.

Fig. 10 shows the comparison of the prediction-map fragmentation of the boundary-related and inference-fusion variants. Lower fragmentation index and small-island ratio values generally indicate fewer fragmented regions and isolated small prediction patches. Removing the edge head (A3), boundary loss (A4), or both components (A5) increases both metrics relative to the full PC-MTPNet, indicating that boundary auxiliary supervision contributes to spatial consistency. In particular, the fragmentation index/small-islandratio values are 0.0456/0.0498 for A3, 0.0346/0.0323 for A4, and 0.0247/0.0324 for A5, compared with 0.0139/0.0199 for the full model. On the contrary, A6 without Hann–Gaussian fusion exhibits a lower fragmentation index of 0.0039 but a substantially higher small-island ratio of 0.0530. Therefore, Hann–Gaussian fusion does not monotonically improve both fragmentation measures; its clearest effect in the present experiment is the suppression of isolated small prediction regions during overlapping-patch fusion. Taken together, the two spatial metrics indicate that the complete PC-MTPNet provides a more balanced full-scene prediction structure.

F. Ablation Study

To quantitatively evaluate the contributions of the major components of PC-MTPNet, we conducted a componentwise removal ablation study. The complete PC-MTPNet is denoted as A0, and six ablation variants, A1–A6, were constructed by disabling different components while keeping the remaining experimental settings unchanged. The detailed component configurations and their absolute quantitative results are summarized in Table V. Checkmarks indicate enabled components, while blank cells indicate disabled components.

In particular, A1 removes MTP-Desc. Because MTP-Polar constructs class prototypes and margin constraints in the MTP-Desc statistical descriptor space, MTP-Polar cannot operate independently when MTP-Desc is absent and is therefore also disabled in A1. Accordingly, A1 represents removal of the complete MTP-Desc-dependent prototype-constraint branch rather than an isolated MTP-Desc-only ablation. A2 retains MTP-Desc but removes MTP-Polar, thereby directly evaluating the contribution of prototype-constrained learning when the statistical descriptor remains available. A3 removes the edge head, A4 removes boundary loss, A5 removes both the edge head and boundary loss, and A6 removes Hann–Gaussian fusion during full-scene inference.

As presented in Table V, the complete PC-MTPNet (A0) achieves an OA of 93.73%, an mIoU of 86.49%, a Macro-F1 of 92.66%, and a Kappa of 92.11%, together with D-J and T-H accuracies of 93.50% and 92.49%, respectively. Its D-J/T-H mean and minimum accuracies reach 93.00%

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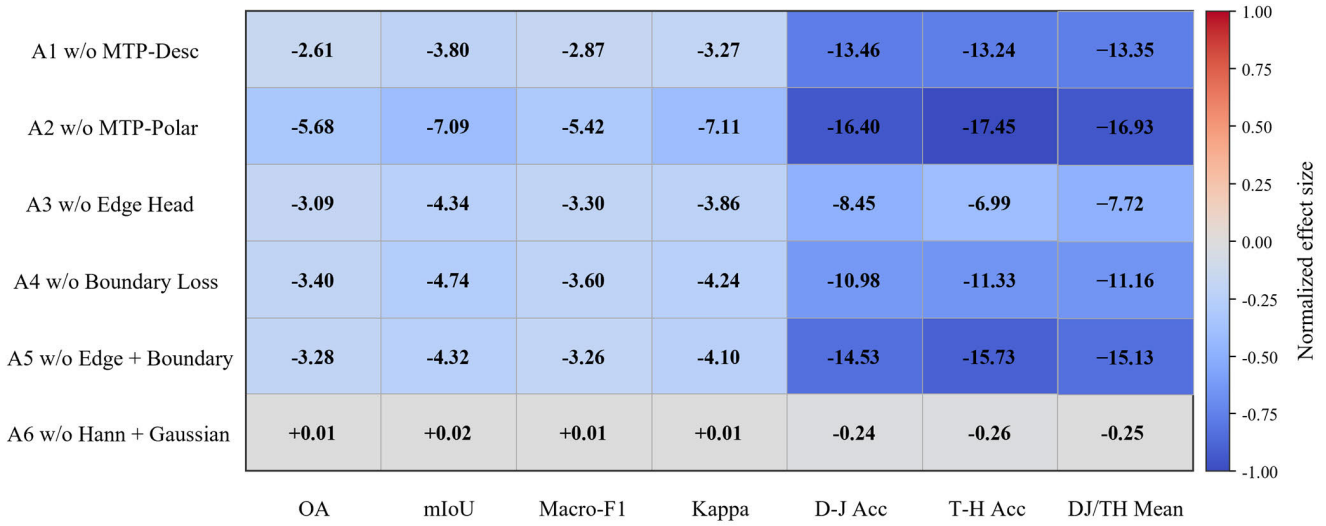


Fig. 10. Normalized ablation effect relative to full PC-MTPNet.

and 92.49%, indicating balanced recognition of both rice subclasses.

Removing the MTP-Desc-dependent prototype branch in A1 decreases OA, mIoU, Macro-F1, and Kappa to 91.12%, 82.69%, 89.79%, and 88.84%, respectively. More pronounced degradation is observed for the two rice subclasses, whose accuracies decrease to 80.04% for D-J and 79.25% for T-H. Because MTP-Polar is necessarily disabled together with MTP-Desc in A1, these changes should be interpreted as the effect of removing the complete statistical descriptor–prototype-constraint pathway rather than the isolated contribution of MTP-Desc.

A2 retains MTP-Desc but removes MTP-Polar and produces the largest overall degradation among the ablation variants, with OA, mIoU, Macro-F1, and Kappa decreasing to 88.05%, 79.40%, 87.24%, and 85.00%, respectively. The D-J and T-H accuracies further decrease to 77.10% and 75.04%, corresponding to reductions of 16.40 and 17.45 percentage points relative to A0. These results demonstrate the central role of MTP-Polar in converting the multi-temporal polarimetric statistical description into an explicit discriminative constraint. The fact that the complete A0 configuration substantially outperforms both A1 and A2 further indicates that MTP-Desc and MTP-Polar are most effective when jointly employed as a descriptor–prototype-constraint mechanism.

The boundary-related variants A3–A5 further demonstrate the contribution of auxiliary boundary supervision. Removing the edge head in A3 decreases mIoU from 86.49% to 82.15%, while D-J and T-H accuracies decrease to 85.05% and 85.50%, respectively. Removing boundary loss in A4 results in an mIoU of 81.75%, with D-J and T-H accuracies of 82.52% and 81.16%. When both the edge head and boundary loss are removed in A5, the D-J and T-H accuracies further decrease to 78.97% and 76.76%, respectively. Although A5 does not produce the largest decrease in every aggregate metric, it causes the strongest rice-subclass degradation among the boundary-related variants. Together with the fragmentation results in Fig. 10, these observations indicate that the edge head and boundary loss provide complementary supervision that is particularly important for maintaining stable D-J/T-H discrimination and spatial coherence along complex SAR boundaries.

On the contrary, removing Hann–Gaussian fusion in A6 produces negligible changes in the aggregate pixel-level metrics: OA, mIoU, Macro-F1, and Kappa change by only +0.01, +0.02, +0.01, and +0.01 percentage points relative to A0, respectively. Meanwhile, D-J and T-H accuracies decrease slightly by 0.24 and 0.26 percentage points. This behavior is consistent with the role of Hann–Gaussian fusion as an inference-stage spatial integration strategy rather than a feature-learning component. As shown in Fig. 10, A6 reduces the Fragmentation Index from 0.0139 to 0.0039 but increases the small-island ratio from 0.0199 to 0.0530. Therefore, Hann–Gaussian fusion should not be interpreted as uniformly improving all pixel-level or fragmentation metrics. Instead, its most evident contribution is the suppression of isolated small prediction regions and the stabilization of overlapping-patch fusion, while having little influence on aggregate classification accuracy.

Overall, the componentwise ablation results clarify the complementary roles of the proposed modules. MTP-Desc provides the physically interpretable statistical space required by the prototype constraint, whereas MTP-Polar plays a central role in enhancing the discriminative organization of D-J and T-H within this space. The edge head and boundary loss improve rice-subclass stability and spatial boundary consistency, while Hann–Gaussian fusion mainly affects the spatial organization of full-scene inference rather than aggregate pixel-level accuracy. These results jointly validate the design rationale of PC-MTPNet.

G. Computational Complexity Analysis

To further evaluate the computational cost of the proposed method, we compared PC-MTPNet with representative baseline methods in terms of trainable parameters, FLOPs, inference latency, and peak GPU memory. The parameter

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TABLE VI
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT METHODS

Method	Params (M)	FLOPs (G)	Inference Time (ms)	Peak Memory (MB)	OA (%)	mIoU (%)
PC-MTPNet (Ours)	**52.49**	**201.26**	51.90	670.20	**93.73**	**86.49**
CVNN-PolSAR	42.11	158.03	**12.86**	704.85	90.68	81.38
SegFormer-B1	23.82	108.76	19.72	374.47	89.71	79.16
U-Net++	26.22	93.30	19.89	325.50	81.93	67.09
SegFormer-B0	3.79	49.54	20.03	**133.78**	70.15	51.88

counts and FLOPs are calculated from the corresponding model structures.

As presented in Table VI, PC-MTPNet has 52.49-M parameters and 201.26-G FLOPs. Although it introduces additional multi-temporal polarimetric descriptor and prototype-constraint components, it achieves the best overall performance among the compared methods, with an OA of 93.73% and an mIoU of 86.49%. Compared with SegFormer-B1 and U-Net++, PC-MTPNet requires more computation but provides higher segmentation accuracy and stronger fine-grained rice discrimination. This indicates that the additional computational cost is justified by the improved classification performance.

V. DISCUSSION

A. Role of Multi-Temporal Polarimetric Statistical Description in Fine-Grained Rice Classification

The comparative analysis in Section IV shows that pixel-level classification models alone are insufficient to distinguish D-J and T-H rice types effectively. At a single-temporal phase, the two rice types often exhibit similar backscattering responses, while their differences are more clearly reflected in cross-temporal variations, channel-energy distribution, channel dispersion, and polarimetric statistical structure. The MTP-Desc module extracts statistical features such as scattering intensity, channel dispersion, entropy, and adjacent-phase variation from multi-temporal polarimetric data, and summarizes them into a physically interpretable descriptor space. This statistical description does not replace the original 48-channel input, but provides a physically meaningful constraint basis for subsequent prototype regularization. The improvements of PC-MTPNet in D-J/T-H recognition and overall metrics indicate that multi-temporal polarimetric statistical description helps alleviate confusion between the two rice types, further supporting the importance of multi-temporal information for fine-grained rice classification.

B. Significance of Prototype Constraints for Reducing Rice-Class Confusion

Conventional pixelwise softmax loss mainly optimizes single-pixel label prediction. When class distributions are similar, the model can easily produce confusion in the probability space. The MTP-Polar module constrains class prototypes in the statistical space constructed by MTP-Desc. In particular, prototype alignment enhances intraclass consistency, while D-J/T-H margin separation increases the statistical distance between the two rice types. This prototype constraint improves probability organization in both probability heatmaps and confusion structures, thereby helping reduce confusion between rice classes.

C. Boundary Stability and Full-Scene Mapping Consistency

SAR imagery is affected by speckle noise, mixed pixels, and local scattering anomalies, which can lead to fragmented boundaries in prediction maps. To improve boundary stability, PC-MTPNet introduces an edge head and boundary loss during training to enhance boundary supervision, and applies Hann–Gaussian sliding-window fusion during inference to reduce probability discontinuities near patch edges. The fragmentation analysis in Fig. 10 shows that boundary supervision consistently improves both spatial metrics, whereas Hann–Gaussian fusion primarily reduces the small-island ratio. Together, these components provide a more balanced full-scene spatial structure.

VI. CONCLUSION

This study focuses on fine-grained rice classification under complex polarimetric SAR backgrounds, with the core task of distinguishing D-J from T-H, rather than performing general land-cover classification. The proposed PC-MTPNet takes multi-temporal polarimetric SAR data comprising six temporal phases and eight channels per phase as input. It first constructs a physically interpretable MTP-Desc space through MTP-Desc to explicitly extract scattering intensity, channel dispersion, channel entropy, and temporal variation features. MTP-Polar is then introduced in this statistical space to impose class prototype constraints and D-J/T-H margin separation, thereby enhancing intraclass consistency and interclass separability. To further improve boundary representation and full-scene inference stability, the network incorporates an edge head and boundary loss during training, and applies Hann–Gaussian sliding-window fusion to improve full-scene prediction stability during inference.

PC-MTPNet achieves an OA, mIoU, Macro-F1, and Kappa of 93.73%, 86.49%, 92.66%, and 92.11%, respectively. The recognition accuracies for the rice subclasses D-J and T-H reach 93.50% and 92.49%, respectively. Compared with the baseline U-Net++, PC-MTPNet improves OA, mIoU, Macro-F1, and Kappa by approximately 11.80, 19.40, 12.83, and 14.72 percentage points, respectively, and improves D-J accuracy and T-H accuracy by approximately 17.57 and

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16.50 percentage points, respectively. These results demonstrate the effectiveness of PC-MTPNet for fine-grained rice classification and support the role of multi-temporal polarimetric statistical description and prototype constraints in alleviating D-J/T-H confusion. Meanwhile, the recognition performance for complex background classes such as water, shoal, and urban remains stable. Boundary supervision and Hann–Gaussian fusion further improve spatial continuity and full-scene mapping stability, resulting in more coherent prediction maps.

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